Questions on the remaining value-group profile switch from Theorem 3.11 to Corollary 3.12

Small questions bearing on the central residual in one route to Corollary 3.12 and ABC: globalization of local shears and object membership

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Abstract

In a conditional assembly from Theorem 3.11 to Corollary 3.12 in inter-universal Teichmüller theory, the chart calibration, normalized-generator scale and unit side can be treated as supplied at the level of stated inputs, while a value-group profile switch H*val remains. A profile-level countermodel shows that this input does not follow from the remaining assembly hypotheses. This note does not claim to solve that residual. Instead, it decomposes it into small questions phrased, as far as possible, in the common vocabulary of anabelian geometry, and records the points on which expert guidance would be decisive.

The central issue is local-to-global: how does a local shear, that is, a freedom arising from a choice of splitting, model or identification, act on degree readouts, and can such a difference be supported at a single bad place as the local component of a global change of model? Around this main question we separate the category in which degree-moving shears should be classified, the minimal category containing both theta sections and degree readouts, the existence of a global difference δ, a finite Tate-curve computation, and two return bridges to the assembly: identification of the two-path difference with the [EtTh] section difference, and passage from equality of readouts to object-level membership.

For each question we record its relation to Project LANA's (9-1). This is a division-of-roles map, not a comparison of progress and not an alternative to (9-1). Some questions would remain meaningful after (9-1) is settled, while answers to others could clarify where any route to (9-1) must carry its load.

The abstract implication cores used in the reduction map have been checked in Lean: differences of two splittings, the volume implication from object membership and monotonicity, finite averages, non-cancellation for a common ratio, and many-to-oneness of totals as shadows of profiles. The questions themselves, and their identification with the actual [EtTh] and IUT constructions, are not formalized.

Keywords: inter-universal Teichmüller theory, Corollary 3.12, value-group profile, local shear, global synchronization, mono-theta environment, anabelian geometry, Lean.

Position of this note

This note is a problem statement for joint examination, not a completed proof. The questions are not addressed to any particular individual or project; they are addressed broadly to anyone with expertise in the relevant areas who may be able to supply references, corrections, counterexamples, or a better formulation. It does not claim a proof or disproof of Corollary 3.12 or the ABC conjecture, a determination of H*val, or an equivalence between Statement P′ below and the residual. It is also not an alternative to, or a criticism of, Project LANA's (9-1). The intended relation is complementary: (9-1) fixes an object-level membership statement on the source side, whereas this note separates definitions, local mechanisms and globalization steps that precede or surround such a statement. Even if every question here receives a positive answer, the two return bridges and the numbered IUT inputs remain separate. Conversely, a concrete countermodel to one question would locate a place where this route closes.

We use the following labels for relations with (9-1): *directly connected* under a fixed dictionary; *connected under an additional identification*; *requires separate examination*; and *corresponding*, meaning essentially the same content in different vocabulary under a fixed dictionary. Here "dictionary" means a fixed correspondence between the assembly-side two paths, possible images and degree data and the membership formulation used by LANA. The dictionary itself is not proved in this note.

1 Fixed background

1.1 The residual in two-path form

The residual in the conditional assembly [U26c] compares two routes at the level of objects. Route 1 starts from the Θ-pilot, passes through the value-group part of one Θ_{LGP}-link to the q-pilot, and then uses the column-one log-Kummer correspondence. Route 2 starts from the same Θ-pilot, uses the column-zero log-Kummer correspondence and permutation symmetry to obtain a possible image. The object-level assertion (α) says that the Route 1 representation is, up to the permitted indeterminacies, one of the Route 2 possible images. The numerical assertion (β) follows from (α) together with the decomposition identity expressing the q-side readout as the volume of that object and monotonicity of the hull volume. We use (α) as the standard form of the residual.

1.2 Parts treated as supplied

The questions below fix the following layers as supplied at assembly-statement level. This is a focusing device, not a claim that they have been proved unconditionally in the actual theory.

Layer	Content	Status here
Chart	Calibration, generator scale and transport distortion of the measuring chart	Treated as supplied
Unit	Theorem 3.11(iii)(a)(b), Kummer classes of actual functions and the restriction bridge	Treated as supplied
Semantics	Reading prime-strip morphisms as isomorphisms of collections of data	Fixed as a reading; actual inhabitation remains separate
Profile	Hull compatibility of the switch $\{j^2u\}_j$ to $\{u\}_j$	**Open; subject of this note**

1.3 What the profile countermodel fixed

The countermodel of [U26d] uses concrete data at $\ell = 7$ in which chart calibration, the normalized generator, the unit side, membership and both volume decompositions hold, while only the labelwise profile differs and the Corollary-3.12-shaped conclusion fails. The aggregate scale is in general $12/(\ell(\ell+1))$. Hence an input solving the residual cannot be merely the numerical assumption that an aggregate scale equals 1; it must connect profiles at object level or at the level of a sufficiently rich readout.

1.4 Nearby public Lean work

The public Project LANA Lean development already contains formalizations close in theme, including abstract forms of [EtTh] Proposition 2.14, shift automorphisms, discrete and cyclotomic rigidity, and, in Wall/LogLink models, scaling and unit-induced properties, a profile coefficient `sFactor`, and hexagon noncommutativity. This note does not restate those developments. Its questions concern how such model-level classifications connect with the categories of the source texts and with globalization of section differences.

1.5 Formal cores checked in Lean, and what was not checked

Before circulation, the parts of the implication map that can be stated without the [EtTh] or IUT constructions were isolated in the Lean module:

`M3.relative.ActualValuePartQuestionImplications`.

A selected build over three modules passed, and `#print axioms` for the main theorems used only the standard three axioms. The checked abstract implications are:

- two splittings of one group homomorphism agree on the quotient, their pointwise difference lies in the kernel and satisfies the usual cocycle identity; hence any readout factoring through that quotient agrees for the two splittings;
- object membership, the readout decomposition identity, the hull-domain condition and monotonicity imply the numerical volume bound, abstractly $(\alpha) \Rightarrow (\beta)$;
- equality of an otherwise unconstrained transported number with the q-side readout does not imply that bound, and equal totals do not determine a labelled profile;
- a normalized linear readout invariant under all label permutations is an average, the mean-square formula holds, and a common ratio $r \neq 1$ at every place does not cancel in a nonzero total.

The Lean files do not formalize mono-theta environments, the actual difference δ, its globalization, Statement P$'$, the two return bridges, or dictionary-dependent relations among candidate forms. They should not be read as verifying the questions themselves.

1.6 Relation to prior public claims, in the audited range

The central identification difficulty itself is not new: it has been a public issue since [SS], and Project LANA records a corresponding membership problem as (9-1). Joshi has published, inside his own arithmetic Teichmüller theory, a claimed local prototype of Corollary 3.12 and valuation-scaling results [JosII]. Because the definitions and framework differ, this note does not treat those claims, whose evaluation is not attempted here, as directly closing Questions Q1 or Q2 as phrased in the

[EtTh] categories. For local outer automorphisms arising from the Jannsen–Wingberg presentation, see [NSW], [HN] and [Kon]; in the statements audited here, the emphasis is on unit/log-side actions, not a category-by-category classification of actions on a normalized degree quotient or a one-place global realization. A public document [Zen26] claims an additional hull-inflation obstruction even under (9-1); that claim is not verified or used here. Accordingly, this note does not claim that Q0–Q7 are globally unprecedented or open in every formulation. In the range audited, no numbered statement was located that closes them in the exact categories and with the exact readouts used below.

2 Setup and terminology

2.1 Stage

Let K be a number field, E/K an elliptic curve, and S a finite set of places at which E has split multiplicative reduction. For $v \in S$, let $q_v \in K_v^\times$ be the Tate parameter, with $\mathrm{ord}_v(q_v) > 0$ for the normalized valuation. Fix a prime $\ell \geq 5$, a bound inherited as a hypothesis from the assembly-side data conditions. For each v, consider the Tate curve, its ℓ-torsion points, the relevant hyperbolic fundamental groups, and decomposition groups at cusps and torsion points. Labels are $j \in \{1, \ldots, \ell^*\}$, $\ell^* = (\ell - 1)/2$. Write $\eta_{v,j}$ for the Kummer class of the theta value. Its valuation satisfies

$$\mathrm{ord}_v(\theta_{v,j}) = \frac{j^2}{2\ell}\,\mathrm{ord}_v(q_v).$$

With $u_v := |\log(\underset{=v}{q})| = (2\ell)^{-1}\log(\mathfrak{q}_v)$, the signed readouts are $-j^2 u_v$ on the theta side and $-u_v$ on the q side.

2.2 Two sections and their difference

A mono-theta environment in [EtTh] contains tempered fundamental group data and a theta section. The cyclotomic envelope also has a tautological algebraic section. In a relevant subquotient the two appear as splittings of one surjection, with the theta section obtained from the algebraic section by subtracting an η-cocycle. We write δ generically for such a difference or for the change of section under a change of model, always naming the category in question. This avoids confusing the standard η-cocycle of [EtTh] with indeterminacy under conjugation or replacement of models.

A mono-theta environment contains the theta section as data; the algebraic section is detected, in subquotients, through commutators. A bi-theta environment contains both sections but loses discrete rigidity. This tension is Question Q3.

2.3 What "shear" means here

Because the word is used with different scopes, we use a working definition. For a replacement α of splittings, models or identifications in a specified category, fixing specified group-theoretic and cyclotomic data, consider the induced action on degree readouts. Conceptually there are two types:

Type	Form of action	Effect on degree
U (splitting shear)	Splitting changes by a unit-valued cocycle	Normally invisible in the valuation quotient
D (degree shear)	Normalized degree readout itself changes non-trivially	If linear, $D \mapsto \lambda D$, $\lambda \neq 1$, is the issue

A warning is essential. Local automorphisms arising from the Jannsen–Wingberg presentation and often called shears mix generators and induce unipotent actions on the unit/log side [NSW, Thm. 7.5.14], [HN], [Kon, Thm. 1.1]. Under suitable hypotheses such an action need not be a scalar composed with a field automorphism, and it may move sizes inside a log shell. This should not be identified with a D-type action on the normalized valuation quotient or degree readout itself. Q0 therefore does not presuppose the existence of D-type shears; it asks for their classification.

When an action on a generator readout is multiplicative, write its ratio as λ_v. Otherwise λ_v is only shorthand and the action itself should be recorded. For transported readouts write $\varepsilon_v := D_v^{\mathrm{tr}}(q_v) - D_v(q_v)$.

3 The local skeleton already in [EtTh]

This section records a starting point, not a new result. The theta functional equation [EtTh, Prop. 1.4(ii)] and its cohomological form [EtTh, Prop. 1.5(iii)] are the source of the j^2 profile. By [EtTh, Thm. 1.6(ii)], a tempered isomorphism preserves the kernel in the valuation chain and the element $1 \in \widehat{\mathbb{Z}}$; by (iii), the $\mathcal{O}^\times \cdot \eta^\Theta$ class is preserved up to conjugation. Thus the indeterminacy of the theta class is compressed to the unit side, while its value part is pinned. Corollaries 2.8(iii) and 2.9 preserve the glued class without a constant indeterminacy and preserve cusp labels. Proposition 2.12 detects the algebraic splitting through commutators; Proposition 2.14(ii) gives a shift automorphism inducing identity on both quotient and kernel; and Corollary 2.19(i) obtains a cyclotomic isomorphism from the difference of two splittings.

Formally, a readout factoring through the common quotient takes the same value for the two splittings. This abstract part is among the Lean-checked cores. It is only readout invariance, not object identity, and it does not include identification of the actual degree readout with such a quotient readout, globalization, or transport to the procession. The questions below separate those outer steps.

4 Small questions Q0–Q7

None of the following questions by itself asserts Corollary 3.12 or ABC. For each one, a known reference, a countermodel, a correction of category, or a better formulation would serve the purpose of this note. Structurally, Q0–Q5 cut out prerequisites, definitions and the local core of Statement P′ in Section 5. Q6 and Q7 are bridges back from P′ to the assembly. Thus P′ is not an additional independent question beside Q0–Q7; it is a composite form built from Q0–Q5, with Q6–Q7 outside it.

Q0. In which category do local shears live?

For a mono-theta environment over a local field and a degree readout D, classify the actions on D induced by automorphisms or changes of model fixing specified group-theoretic and cyclotomic data. In particular, in which of the following categories does a D-type action, moving degree itself, occur?

1. topological-group automorphisms of tempered fundamental groups, modulo inner automorphisms;
2. automorphisms of mono-theta environments;
3. automorphisms of bi-theta environments;

4. automorphisms of tempered Frobenioids and their splitting monoids;

5. automorphisms of realified Frobenioids.

Please also distinguish whether "fixing cyclotomic data" means identity action on the cyclotomic object or compatibility with its natural cyclotomic isomorphism.

Relation to (9-1): requires separate examination. (9-1) concerns membership of specific standard objects and does not classify automorphisms category by category. Even if (9-1) is true, degree-moving shears might exist in a wider category; even if they exist, the standard transport might avoid them.

Q1. Can a local shear globalize? (Main question)

Fix the local shear objects as in Q0. What is the image, inside the product of local shears over places, of differences arising from a global change of model? When the degree action can be written by ratios $(\lambda_v)_{v\in S}$, can the family

$$\lambda_{v_0} \neq 1 \quad \text{at one bad place } v_0 \text{ only}, \qquad \lambda_v = 1 \quad (v \neq v_0)$$

be realized? If not, is the obstruction only the product formula, label synchronization, rigidity of a decomposition-group preserving global model, or another global constraint? We ask for the distinction between a purely global anabelian reading, using arithmetic fundamental-group data and decomposition groups, and a synchronized reading in which label synchronization of the type in [IUTchI, Prop. 6.5] is part of the given data.

If a one-place shear is realizable but compensated at other places so that the total is preserved, the form of compensation should be made explicit. Formally, a common ratio $r \neq 1$ at all places does not cancel in a nonzero total; what remains open is whether place-dependent families, and one-place support in particular, occur in the image of global differences.

Relation to (9-1): requires separate examination. (9-1) states membership for standard global objects; it does not classify realizability of prescribed local families. Under a fixed dictionary and Bridge (a), total consequences for the standard difference may follow, but not the one-place support condition.

Q2. Is the degree readout invariant under the standard difference of sections?

For the algebraic and theta sections in [EtTh], if the identification of coefficients is transported to the theta-section side, is the valuation-quotient readout of the q-parameter Kummer class invariant? Precisely, does $\varepsilon_v = D_v^{\mathrm{tr}}(q_v) - D_v(q_v)$ vanish? Is this already a direct consequence of Propositions 2.12 and 2.14(ii), Theorem 1.6(ii)(iii), and Corollary 2.19(i), or is an additional claim needed at the coefficient-identification step? Abstractly, invariance is immediate once the readout factors through a quotient on which the two splittings agree. The remaining issue is the source-level identification showing that the actual degree readout and transport have that form.

Relation to (9-1): connected under an additional identification. With the assembly dictionary and readout identification fixed, a numerical shadow follows from (9-1); the exact transformation rule inside [EtTh] does not. This question may close from existing [EtTh] results alone.

Q3. What is the smallest category containing both sections and the degree readout?

What is the smallest category in which the algebraic section, the theta section and the degree readout simultaneously exist as data while discrete rigidity is retained? In a mono-theta environment

the algebraic section is only detected in subquotients through commutators, whereas a bi-theta environment has both sections but loses discrete rigidity. Is there a standard category resolving this tension, or evidence that the tension itself is essential?

Relation to (9-1): requires separate examination. Even if (9-1) is formulated in the LANA framework, that does not decide which mono/bi-theta environment is the minimal home for this question. It is a question at the level of source definitions.

Q4. Does a global δ exist without Hodge theatres?

Can the projective system of local mono-theta environments and its η-classes be glued, from global arithmetic fundamental-group data and restrictions to decomposition groups alone, into one global object δ whose local components are not independent? If so, what functor or reconstruction algorithm gives the gluing? If not, what is the smallest additional datum required? Is the understanding correct that label synchronization alone does not give cocycle coherence?

Relation to (9-1): requires separate examination. A global object in the LANA framework would still need to be identified with a gluing of [EtTh] section systems. Whether the setup can be founded in pre-IUT vocabulary is not decided by granting (9-1).

Q5. Can the degree action of δ be computed in a Tate prototype?

For a Tate curve over a finite p-adic field containing the necessary roots of unity, for example at $\ell = 5$ and including Gaussian-type local test cases, make explicit from theta series and product expansions the classes $\theta_{v,j}$ and the change of η-classes under translation by $a \in \mathbb{Z}$. Under the standard splitting obtained by restriction to the zero section, compute the transported degree action ε_v, separating classically known parts from parts depending on the choice of section.

Relation to (9-1): partly known; the centre requires separate examination. Valuations of theta values and the functional equation follow from classical computation and [EtTh, Props. 1.4–1.5]. The value of ε_v under the standard splitting is not computed by (9-1).

Q6. Bridge (a): is the assembly two-path difference the [EtTh] section difference?

At each bad place, restrict the composite "Route 1 theta-link value part $\circ$ (Route 2 permutation transport)$^{-1}$" to the data of the mono-theta splitting monoid of [EtTh, Thm. 2.2(ii)]. Does it agree with transport induced by the [EtTh] section difference, the η-cocycle? Is that agreement compatible, including degree readouts, with the passage from the single-basepoint method to the procession method in [IUTchII, Rem. 3.5.3] and with horizontal compatibility in [IUTchIII, Cor. 2.3(iv)]? If this is immediate from a numbered statement, we would be grateful for that reference.

Relation to (9-1): corresponding under a fixed dictionary. With a dictionary between LANA objects and the assembly two paths, a positive answer largely overlaps the content of (9-1). But fixing that dictionary is itself an unproved bridge, and identification with the [EtTh] δ is not immediate from the statement of (9-1).

Q7. Bridge (b): what is needed to pass from equal readouts to object membership?

Under a splitting into unit part and a rational ray generated by a normalized element, can two pilot objects with the same unit part and the same total procession degree be treated as the same possible image in the hull-compatible sense? If not, what is the smallest additional datum beyond

total degree: the profile tuple itself, a lower-bound or inclusion condition for regions, or another object-level identification?

Relation to (9-1): membership is supplied; the implication requires separate examination. Granting (9-1) supplies the needed membership itself. It does not prove an implication "equal readouts $\Rightarrow$ membership". A total degree is a many-to-one shadow of a tuple, and without additional structure such an implication fails in general. In the assembly-shaped countermodel, this insufficiency is visible when the transported number is not connected to a theta-side object.

5 Statement P′

Statement P′ is a revised presentation of an earlier reformulation, arranged to match the local skeleton and the decomposition into questions above. It is not claimed to be equivalent to the residual. P′ is not an additional independent problem beside Q0–Q7: Q0–Q5 separate its prerequisites, definitions and local core, while Q6–Q7 are bridges outside P′ leading back to the assembly.

Part of P′	Questions	Remark
Two sections and δ setup	Q3	Minimal category
Replacement by degree readouts	Q0, Q2, Q5	Shear classification, transported readout, computation
Globalization	Q1, Q4	Image of global differences; existence of global δ
Total compatibility assertion	Q1 standard case + Q2	Standard difference, not arbitrary shear families
Return to assembly	Q6, Q7	Bridges (a)(b), outside P′

5.1 Differences from a solved form

[EtTh, Cor. 2.19(i)] says that the difference of two sections gives a compatible isomorphism on cyclotomic objects. P′ differs in two ways: first, it replaces the object of compatibility by value-group degree readouts; second, it states the local theory globally, under one global δ and the product-formula constraint. The geometric Szpiro inequality in the function-field case is only an expository anchor for the second difference: there globality appears as a closed intersection-theoretic total. It does not provide a mechanism in the number-field case. For a prior formulation of a geometric Corollary 3.12, see [Jos].

5.2 Setup and statement

Take K, E/K, S, q_v, $\ell \geq 5$, theta torsion data and $\eta_{v,j}$ as in Section 2. Take the two sections s^{alg}, s^{Θ} and their difference δ in the sense of Section 2; cyclotomic compatibility is known by [EtTh, Cor. 2.19(i)]. As a marked **premise-or-part-of-the-question**, suppose that the sections and δ exist as one global object whose local sections are restrictions to decomposition groups. This globalization is not founded here from the local [EtTh] theory; it is Q4.

Statement 1 (P′: compatibility of transported degree readouts)**.** δ is compatible with value-group degree readouts. Precisely, for the system over all $v \in S$, the degree datum $\mathrm{ord}_v(q_v)$ read through the algebraic section agrees, as the same readout in the same reconstructed value group, with the degree datum of the same object read through the theta section, that is, from the system of Kummer classes $\{\eta_{v,j}\}$.

What is read through the theta section is not the total degree of the theta system itself, which is computed from its own valuations and does not equal the q-degree. It is the degree of q-data transported by δ. Nor is the assertion about capsulewise profiles $\{j^2\}$ versus $\{1\}$, or a place-by-place assertion. It is asserted only for the system over S, for one global δ, under the product-formula constraint.

Relation to (9-1): connected under additional identifications. With dictionary and Bridges (a)(b) fixed, the numerical content of P′ is obtained as a shadow of (9-1). The converse does not hold: P′ is a readout assertion, not object membership itself.

5.3 Rating of P′

No equivalence between P′ and (α) has been proved; the gap consists of Q6 and Q7. The globalization in the setup is an unverified seam, and a version with the global clause removed is not P′. The local skeleton gives formal scaffolding, but transplantation to degree readouts is itself the open content and does not follow automatically from Corollary 2.19(i), whose proof uses cyclotomic Kummer structure. The direction checked formally is (α) plus decomposition identities implying (β); numerical compatibility P′ alone does not imply (α). If P′ is false, this assembly route closes, but that would not disprove ABC. If P′ is true, Bridges (a)(b) and the IUT black box still remain.

6 Candidate forms A–D and the implication map

Before this question list, the residual (α) was examined in the following candidate forms. We record only their relation to the questions here. The implications are sketches under a fixed dictionary, not proved equivalences between autonomous statements.

Form	Content	Relation here
A	Commutativity, in the value component, of the two-path square	Assembly-side form of (α); Q6 asks for [EtTh]-side identification
B	Application of semantic principle S to the relevant isomorphism and value component	Semantic clothing of A; the general principle is not the question here
C	Commutativity of mono-anabelian reconstruction R with theta transport	Algorithmic clothing of A; seam: assembly degree data are output of R
D = (9-1)	Membership form of $\eta^q = \eta_S^{\mathrm{anab}}$ type	Current reference form; this note asks before and around it
P′	Global compatibility of transported degree readouts	Common-vocabulary entrance; weaker readout form than A

Under supplied inputs, the sketch is: the relevant instance of B corresponds to A and to (α); C would give A through its seam; and D corresponds to (α) under the fixed dictionary. These dictionary-dependent relations were not formalized in Lean. Q0–Q5 separate prerequisites and definitions outside this map; Q6–Q7 ask about its seams.

7 Summary of relations with (9-1)

The labels below mean the position of a question if (9-1) is granted. They do not include the truth of (9-1) or correctness of a dictionary. This is a division-of-roles summary. Many questions retain independent meaning after (9-1), as category classification or foundations for globalization.

Question	Relation	Reason in brief
Q0 local-shear classification	Separate	(9-1) gives no categorywise automorphism classification
Q1 globalization / one place	Separate	Standard membership does not classify prescribed families
Q2 transported-readout invariance	Additional identification	Numerical shadow follows; internal [EtTh] rule is separate
Q3 minimal category	Separate	Source-definition question
Q4 global δ	Separate	Identification with [EtTh] gluing is additional
Q5 Tate computation	Partial	Theta valuations known; ε_v is not computed by (9-1)
Q6 Bridge (a)	Corresponding under dictionary	Dictionary itself unproved
Q7 Bridge (b)	Membership supplied; implication separate	(9-1) does not prove readout-to-membership
P′	Additional identifications	Shadow under dictionary and bridges; converse fails
Forms A, D	Corresponding under dictionary	Rephrasings of (α)
Form B, general principle	Separate	One instance does not imply a general semantic principle
Form C	Additional identification	Assembly degree data as output of R is the seam

8 Closing: what guidance would be decisive?

In the author's understanding, the centre is how far local freedom is killed by global synchronization. This appears most sharply in Q1's image problem. But Q1 is meaningful only after Q0 fixes the category and type of shears and Q3–Q4 fix the home of global objects. That is why the questions are separated.

Any of the following would be decisive for this route:

- for Q0, the exact category in which degree-moving shears exist, with a classification reference;
- for Q1, a one-place-supported global realization, or an explicit constraint forbidding it;
- for Q2 and Q5, a derivation of transported-readout invariance from existing theorems, or a concrete computed value;
- for Q3 and Q4, a standard category containing both sections and readouts, or an existing reconstruction giving global gluing;
- for Q6 and Q7, a numbered statement closing a bridge, or a concrete step where a bridge fails;
- a prior public formulation of the residual in [EtTh] or anabelian reconstruction vocabulary. This note does not make its value depend on a novelty claim.

None of these questions is Corollary 3.12 or ABC itself. However, the conditional assembly and the profile countermodel have localized the remaining difficulty along this route to finitely many steps. The purpose of this note is to put those steps in a form that specialists can judge in everyday vocabulary.

Residual	Content	Treatment here
R1	Numbered statements of IUT I–IV as black boxes	Not verified
R2V	Value-group profile switch H*val	Decomposed into Q0–Q7 and P′
R2	Inhabitation / frame fidelity to the actual construction	Marked at Q6 and Form B seams
R3	Source checking and independent reverification	Author checking and outside rereading remain

Disclosure

This is personal work by an independent researcher, unrelated to any institutional affiliation. Generative AI was used under the author's direction for auditing, formalization and drafting. Responsibility for the contents rests with the author. This is a pre-review work report.

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